YISHUN INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
**Higher 1**

| | |
|---|---|
| CANDIDATE NAME | |
| CG | | INDEX NO | |

# CHEMISTRY 8873/02

Paper 2 Structured Questions **30 August 2021**

Candidates answer on the Question Paper. **2 hours**

Additional Materials: Data Booklet

## READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, and glue or correction fluid/tape.

**Section A**
Answer **all** questions.

**Section B**
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **Paper 1** | |
| | **/30** |
| **Paper 2** | |
| **1** | **/16** |
| **2** | **/10** |
| **3** | **/14** |
| **4** | **/6** |
| **5** | **/14** |
| **6 or 7** | **/20** |
| **Penalty** | |
| | **/80** |
| **Overall (Paper 1 & 2) Percentage (%)** | |
| | **/110** |

This document consists of **21** printed pages and **3** blank pages.

## Section A

Answer **all** the questions in this section, in the spaces provided.

**1** **(a)** At 573K phosphorus(V) chloride, $PCl_5(g)$, decomposes to form phosphorus(III) chloride, $PCl_3(g)$, and chlorine, $Cl_2(g)$. A dynamic equilibrium is established as shown.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ numerical value of $K_c = 0.625$

**(i)** Explain what is meant by the term *dynamic equilibrium*.

…………………………………………………………………………………………

…………………………………………………………………………………………

……………………………………………………………………………………[1]

**(ii)** Suggest and explain **two** changes which will increase the decomposition of $PCl_5$.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

……………………………………………………………………………………[2]

**(iii)** The experiment was repeated at 473K. The numerical value of $K_c$ was found to be $8.3 \times 10^{-3}$. Deduce whether the decomposition reaction is exothermic or endothermic.

Explain your answer.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

……………………………………………………………………………………[2]

**(iv)** The first experiment at 573K was repeated but the total pressure was doubled.

Predict the effect this would have on the value of $K_c$ at 573K. Explain your answer.

[No calculations are required.]

..........................................................................................................

..........................................................................................................[1]

**(b)** The rate of this reaction was measured at different initial concentrations of the two reagents. The table shows the results obtained.

| experiment | [$CH_3CH_2CHClCH_3$] | [$I^-$] | relative rate |
|---|---|---|---|
| 1 | 0.06 | 0.03 | 3 |
| 2 | 0.10 | 0.03 | 5 |
| 3 | 0.06 | 0.05 | 5 |
| 4 | 0.08 | 0.04 | |

**(i)** Deduce the order of reaction with respect to each of [$CH_3CH_2CHClCH_3$] and [$I^-$]. Explain your reasoning.

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................[2]

**(ii)** Write the rate equation for this reaction, stating the units of the rate constant, *k*.

..........................................................................................................

..........................................................................................................[2]

**(iii)** Calculate the relative rate for experiment 4.

[1]

**(c)** **(i)** An important reaction of $CHCl_3$(g) is the manufacture of $CHClF_2$(g), using the following reversible reaction.

$$CHCl_3(g) + 2HF(g) \rightleftharpoons CHClF_2(g) + 2HCl(g)$$

Use the data to calculate the enthalpy change of reaction, $\Delta H_r$, for the formation of $CHClF_2$(g) as shown in the equation.

| compound | enthalpy change of formation, $\Delta H_f$ / kJ mol$^{-1}$ |
|---|---|
| $CHCl_3$(g) | –103.2 |
| $CHClF_2$(g) | –482.2 |
| HF(g) | –273.3 |
| HC*l*(g) | –92.3 |

[2]

**(ii)** Calculate the enthalpy change for the forward reaction in the equilibrium above using bond energy values from the *Data Booklet*.

[2]

**(iii)** Explain why the value calculated in **1(c)(ii)** is different from that calculated in **1(c)(i)**.

……………………………………………………………………………………………….…

…………………………………………………………………………..………………………

………………………………………………………………………………………...…... [1]

[Total: 16]

**BLANK PAGE**

**2** An experiment was carried out to determine the percentage of iron in a sample of iron wire. A 3.35 g piece of the wire was reacted with dilute sulfuric acid, in the absence of air, so that all of the iron atoms were converted to iron(II) ions. The resulting solution was made up to 250 $cm^3$.

**(a)** A 25.0 $cm^3$ sample of this solution was acidified and titrated with 0.0250 mol $dm^{-3}$ potassium dichromate(VI). The results are seen below.

| titration number | 1 | 2 |
|---|---|---|
| initial burette reading / $cm^3$ | 0.00 | 0.50 |
| final burette reading / $cm^3$ | 32.10 | 32.50 |
| volume of potassium dichromate(VI) used / $cm^3$ | | |

**(i)** Complete the table above and find an average volume of potassium dichromate(VI) used in the titration. Leave all answers to 2dp. Hence, use the average volume of potassium dichromate to calculate the number of moles of dichromate(VI) ions used in the titration.

[3]

**(ii)** Using data from the *Data Booklet*, complete and balance the ionic equation for the reaction between the dichromate(VI) ions and the iron(II) ions.

$Cr_2O_7^{2-}(aq) + 6Fe^{2+}(aq) +$ ........$H^+(aq) \rightarrow$ ........$Cr^{3+}(aq) + 6Fe^{3+}(aq) +$ ........$H_2O(l)$

[1]

**(iii)** Calculate the mass of iron in the 3.35 g piece of wire.

[3]

**(iv)** Calculate the percentage of iron in the iron wire.

[1]

**(b)** Another student performed the same experiment but he accidentally used a lower concentration of potassium dichromate(VI). Explain the effect, if any, that the error will have on the mass of iron calculated.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………[2]

[Total: 10]

3 **D**, **E**, **F**, and **G** are four consecutive elements in the **fourth** period of the Periodic Table. (The letters are **not** the actual symbols of the elements.)

**D** is a soft, silvery metal with a melting point just above room temperature. Its amphoteric oxide, $\mathbf{D}_2O_3$, has a melting point of 1900 °C and can be formed by heating **D** in oxygen.

**G** is a solid that can exist as several different allotropes, most of which contain $\mathbf{G}_8$ molecules. **G** burns in air to form $\mathbf{G}O_3$ which dissolves in water to form an acidic solution. This solution reacts with sodium hydroxide to form the salt $Na_2\mathbf{G}O_4$.

**(a)** **(i)** Suggest the identities of **D** and **G**.

**D**: …………………………………………

**G**: …………………………………………

[2]

**(ii)** Write equations for the reactions of $\mathbf{D}_2O_3$ with

I hydrochloric acid,

…………………………………………………………………………………………… [1]

II sodium hydroxide.

…………………………………………………………………………………………… [1]

**(iii)** Write an equation for the formation of an acidic solution when $\mathbf{G}O_3$ dissolves in water.

…………………………………………………………………………………………… [1]

**(b)** State and explain the trend in volatility of chlorine, bromine and iodine down the group.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

**(c)** The fifth to eighth ionisation energies of two elements in the third period of the Periodic Table are given. The symbols used for reference are not the actual symbols of the elements.

| | ionisation energies / kJ mol$^{-1}$ | | | |
|---|---|---|---|---|
| | fifth | sixth | seventh | eighth |
| **X** | 7012 | 8496 | 27 107 | 31 671 |
| **Y** | 6542 | 9362 | 11 018 | 33 606 |

**(i)** State and explain the group number of element **Y**.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

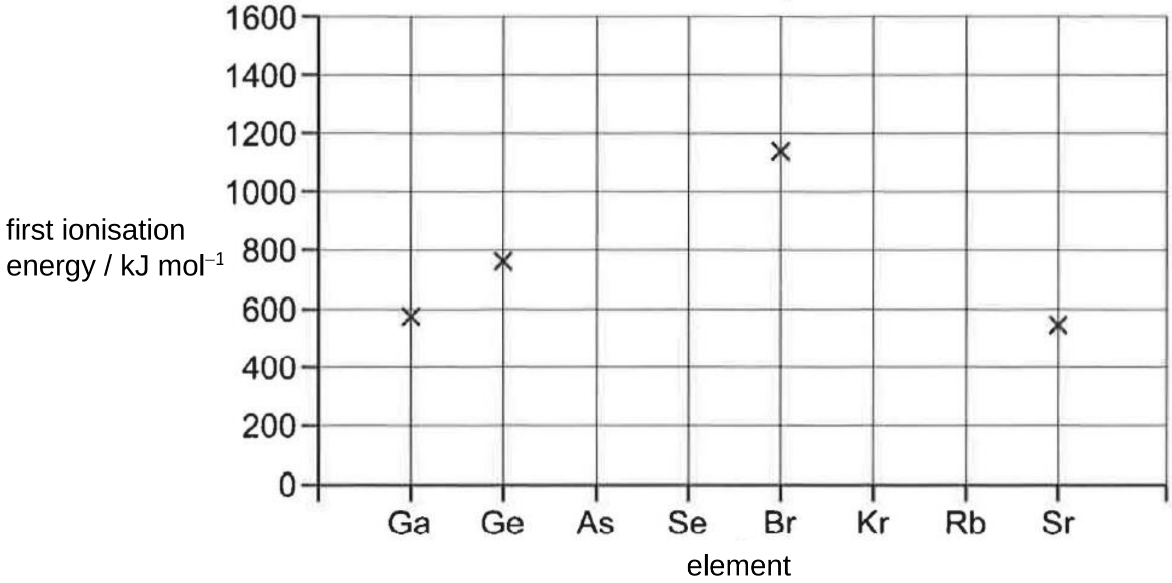
………………………………………………………………………………………………. [2]

**(ii)** Complete the electronic configuration of element **X**.

………………………………………………………………………………………………. [1]

**(d)** The graph below can be used to plot the first ionisation energy of the elements gallium to strontium. Four of these elements have been plotted already.

**(i)** Use your knowledge of the variation in first ionisation energy of the elements of Period 3 (Na to Ar) to estimate and plot, on the graph below, the first ionisation energies of the other four elements: As, Se, Kr and Rb.




[2]

**(ii)** Explain why there is a general increase in the first ionisation energy across the third period.

.....................................................................................................

.....................................................................................................

.....................................................................................................

.....................................................................................................

.....................................................................................................

.....................................................................................................[2]

[Total: 14]

**BLANK PAGE**

**4** Polylactic acid (PLA) is a plastic made from polymerisation of lactic acid, $CH_3CH(OH)COOH$. PLA has become a popular material due to it being economically produced from renewable resources. In 2010, PLA had the second highest consumption volume of any bioplastic of the world. PLA is used for making items as diverse as packaging materials, surgical thread and clothing.

**(a)** **(i)** Draw a repeat unit for PLA and state the type of polymerisation involved.

type of polymerisation: ……………………………………………………………

[2]

**(ii)** Explain why PLA cannot be used to make containers that store strongly alkaline cleaning solutions.

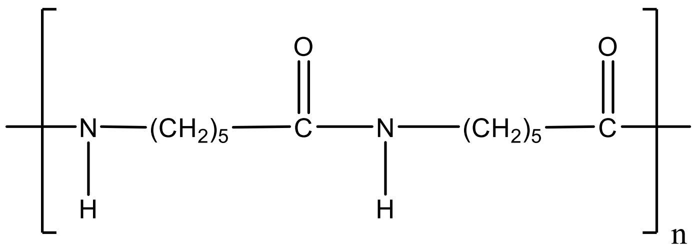
……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………[1]

**(b)** Another polymer that is commonly used in clothing is Nylon 6.

Nylon 6 has the following structure.



Nylon 6

**(i)** Draw the structure of the monomer used to make Nylon 6.

[1]

**(ii)** Explain why clothing made from Nylon 6 is prone to creasing while that made from PLA is wrinkle free.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

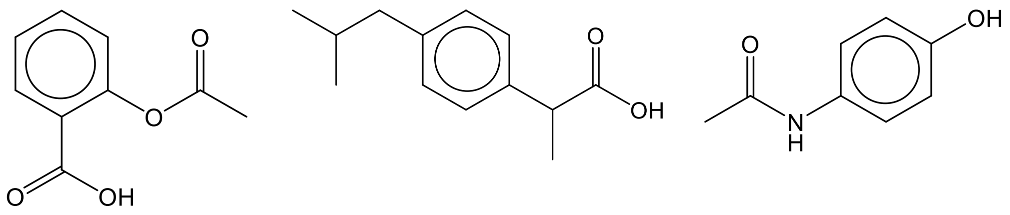
…………………………………………………………………………………………………

…………………………………………………………………………………………………

……………………………………………………………………………………………...[2]

[Total: 6]

**5** **(a)** The pharmaceuticals aspirin, ibuprofen and paracetamol are all painkillers.



aspirin ibuprofen paracetamol

**(i)** Infra-red absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows an absorption at 1710 – 1750 $cm^{-1}$ as shown in the *Data Booklet*.

| Bond | Functional Group | Absorption range / $cm^{-1}$ |
|---|---|---|
| C=O | ester | 1710 – 1750 |
| | | |
| | | |

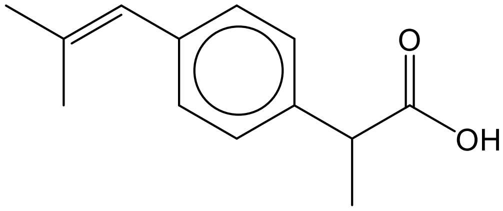
Identify two infra-red absorption ranges that will be shown by aspirin and ibuprofen but **not** by paracetamol using the *Data Booklet*. Fill in your answers by completing the table above. [2]

**(ii)** Paracetamol reacts with hot dilute $H_2SO_4$. Identify the type of reaction which occurred and draw the structures of two organic products formed in the reaction.

Type of reaction:…………………………………………………………………………

[3]

**(iii)** Compound **X** is a derivative of ibuprofen.



**X**

Identify the type of reaction that **X** is able to undergo but not ibuprofen.

……………………………………………………………………………………………… [1]

**(b)** There are four isomeric alcohols with the formula $C_4H_{10}O$.

One of them is butan-1-ol, which can be formed from 1-bromobutane in the laboratory.

**(i)** State and explain whether butan-1-ol or 1-bromobutane has a higher boiling point.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………… [2]

**(ii)** Suggest reagents and conditions to convert 1-bromobutane to butan-1-ol.

………………………………………………………………………………………………

……………………………………………………………………………………………… [1]

**(iii)** Name and draw the structure of the isomeric alcohol which does not react with acidified potassium dichromate(VI).

[2]

**(iv)** All four isomeric alcohols can undergo elimination to form alkenes.

Draw the structure of the alkene formed that exhibits *cis*-*trans* isomerism.

[1]

**(v)** Describe, in terms of orbital overlap, the bonding between the two carbon atoms of the C=C bond in an alkene.

…………………………………………………………………………………………….…

…………………………………………………………………………………………….…

…………………………………………………………………………………………….…

……………………………………………………………………….………………………

………………………………………………………………………………….…...….... [2]

[Total: 14]

**BLANK PAGE**

## Section B

Answer **one** question from this section, in the spaces provided.

**6** **(a)** Define the term *standard enthalpy change of neutralisation.*

.............................................................................................................................

.............................................................................................................................

...........................................................................................................................[1]

**(b)** In an experiment to determine the maximum change in temperature when sodium hydroxide is added to hydrochloric acid, 40 $cm^3$ solution of 2.0 mol $dm^{-3}$ hydrochloric acid is transferred to a polystyrene cup.

The initial temperature of the hydrochloric acid before sodium hydroxide is added is measured and recorded every minute for 3 minutes. At the $4^{th}$ minute, a 20 $cm^3$ solution of 2.0 mol $dm^{-3}$ aqueous sodium hydroxide is poured into the polystyrene cup containing the hydrochloric acid. The temperature of the resulting mixture is then measured every minute for a further 6 minutes.

The results are shown in Fig. 6.1.

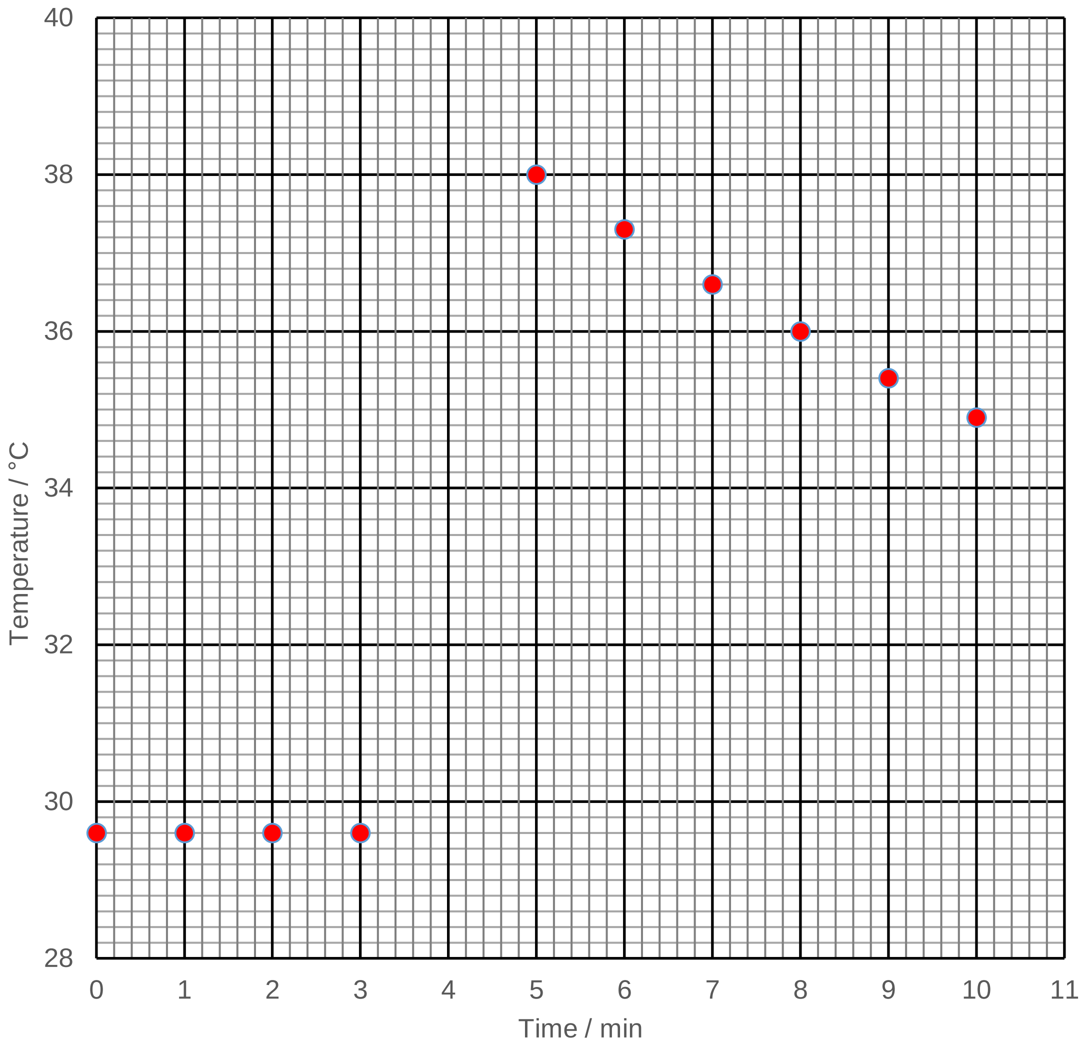



**Fig 6.1**

**(i)** Draw two separate straight lines of best fit. The first should take into account the points before sodium hydroxide was added. The second line should take into account the points after the reaction had finished.

Extrapolate (extend) both lines to four minutes.

Determine a value for the maximum temperature change, $\Delta T_{max}$, at $t$ = 4.0 min using the graph.

Show your working.

$\Delta T_{max}$ = ……………………………………°C
[2]

**(ii)** The specific heat capacity of the mixture can be assumed to be 4.2 J $K^{-1}$ $cm^{-3}$. Using your value in **6(b)(i)**, calculate the enthalpy change of neutralisation of the reaction.

[3]

**(iii)** A student repeated the experiment using 20 $cm^3$ of 2.0 mol $dm^{-3}$ sodium hydroxide and 40 $cm^3$ of 2.0 mol $dm^{-3}$ aqueous ethanoic acid. All other conditions were kept constant.

Suggest whether the temperature change will be more or less than your answer in **6(b)(i)** and give an explanation for your answer.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

**(iv)** Calculate the pH of a $2.5 \times 10^{-4}$ mol dm$^{-3}$ solution of hydrochloric acid.

[1]

**(c)** **(i)** Sketch a Boltzmann distribution curve and use it to explain the effect of a catalyst on the rate of reaction.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[3]

**(ii)** Complete Fig. 6.2 to show how the rate constant varies with concentration, at constant temperature.

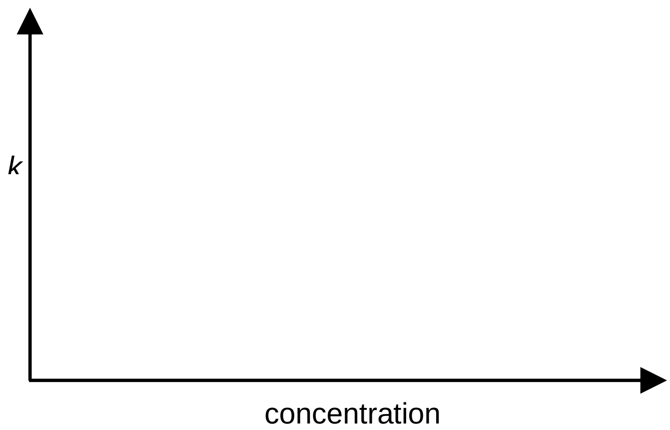



**Fig 6.2**

[1]

**(d)** **(i)** Draw a 'dot-and-cross' diagram for $ClO_2$. (Assume that the halogen atom occupies a central position the molecule.)

[1]

**(ii)** The compound $FO_2$ does not exist, but $ClO_2$ does. Suggest a reason for this difference.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[1]

**(iii)** $ClO_2$ forms the negative ion $ClO_2^-$.

Draw a simple diagram to show how a water molecule can be attracted to a $ClO_2^-$ ion. Label the diagram to show the type of interaction involved.

[1]

**(e)** Silicon carbide is a black solid with a melting point of 2730 °C. It does not conduct electricity in the solid state nor in the liquid state. It is insoluble in water.

Name the type of structure and describe the bonding in the solid silicon carbide.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[2]

**(f)** Propanoic acid undergoes a condensation reaction with ethanol.

Name the product and state the conditions needed for the reaction.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[2]

[Total: 20]

**7** **(a)** Sodium sulfite is a pale yellow solid known for its reducing properties. It is used in the textile industry as a bleaching and desulfurizing agent. Its reducing properties are also exploited in swimming pools in the dechlorination of pool water.

It is made industrially by reacting sulfur dioxide with a solution of sodium carbonate.

$$SO_2 + Na_2CO_3 \rightarrow Na_2SO_3 + CO_2$$

**(i)** Draw the 'dot-and-cross' diagram of $Na_2SO_3$.

[1]

**(ii)** State the shape of the sulfite anion.

…………………………………………………………………………………………...…...[1]

**(iii)** Predict whether $Na_2CO_3$ or $Na_2SO_3$ would have a higher melting point. Explain your answer in terms of structure and bonding.

……………………………………………………………………………………………….…

……………………………………………………………………………..…………………

……………………………………………………………………………………………….…

……………………………………………………………………………..…………………

……………………………………………………………………………..…………………

…………………………………………………………………………………………...….. [2]

**(b)** Aqueous ammonia is a weak base whereas barium hydroxide, $Ba(OH)_2$, is a strong base.

**(i)** Explain the difference between a *weak base* and *strong base*.

……………………………………………………………………………………………….…

……………………………………………………………………………..…………………

……………………………………………………………………………..…………………

…………………………………………………………………………………………...….. [1]

**(ii)** A solution of barium hydroxide has a pH of 12.5. Calculate the concentration of this solution of barium hydroxide.

[2]

**(iii)** Write balanced equations to show how a mixture of aqueous ammonia and ammonium nitrate maintains pH when small amounts of acids and alkalis are added.

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………... [2]

**(c)** Iodine and iodide ions undergo the following equilibrium in aqueous solution.

$$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$$

**(i)** Write an expression for the equilibrium constant for this reaction, $K_c$, stating its units.

[2]

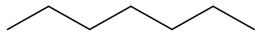
**(ii)** When 2.54 g of solid iodine is dissolved in 100 $cm^3$ of 1.00 mol $dm^{-3}$ KI, and allowed to reach equilibrium, the resulting $[I_3^-(aq)] = 9.98 \times 10^{-2}$ mol $dm^{-3}$.

Calculate the equilibrium concentrations of $[I_2(aq)]$ and $[I^-(aq)]$ in this solution and hence calculate a value for $K_c$.

[3]

**(d)** The polymer, acrylonitrile butadiene styrene, melts on heating. It is used to make sports equipment and musical instruments such as recorders.

Another synthetic polymer, polyurethane, chars (becomes blackened) on heating. It is used to make high-resilience foam seating and rigid insulation panels.

Sketch the structure of each polymer using  as a simplified representation of the polymer chain. Include labels on your sketch. Classify each polymer and use its structure to explain its properties.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [6]

[Total: 20]